

Recovering Top-Two Answers and Confusion Probability in Multi-Choice Crowdsourcing

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Crowdsourcing

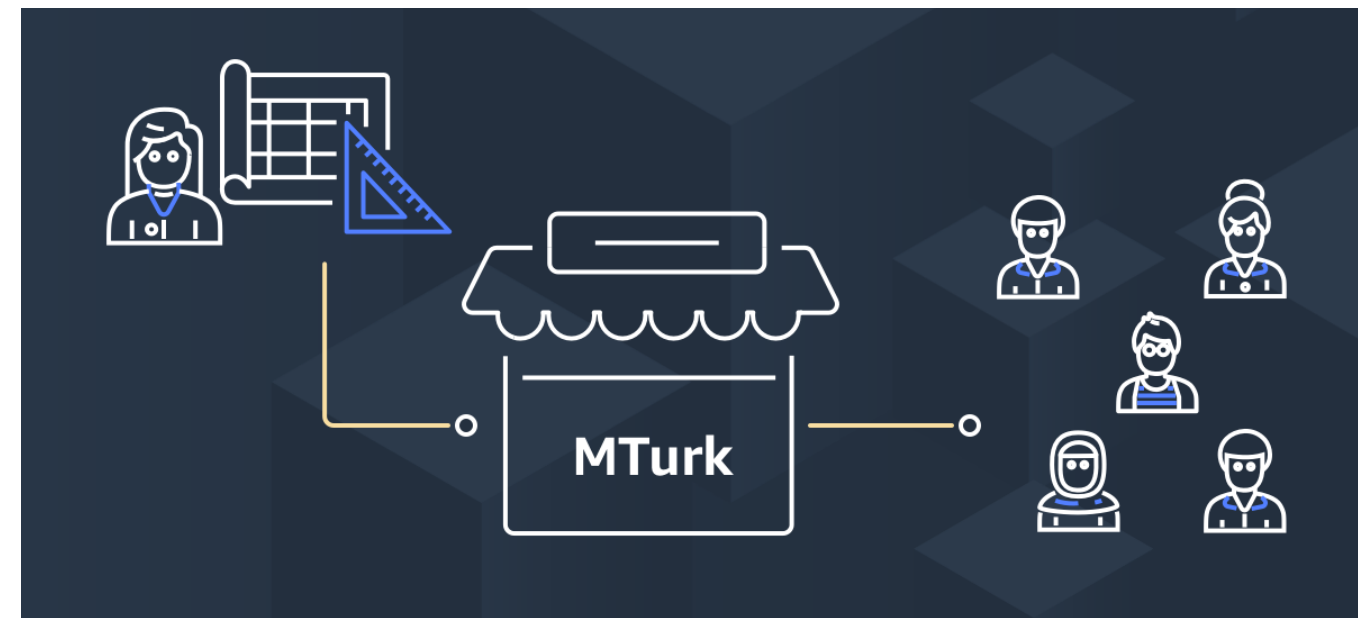


Figure 1. Illustration of Crowdsourcing

- Instead of asking one expert, ask multiple non-experts.
- Efficient in terms of time and cost.

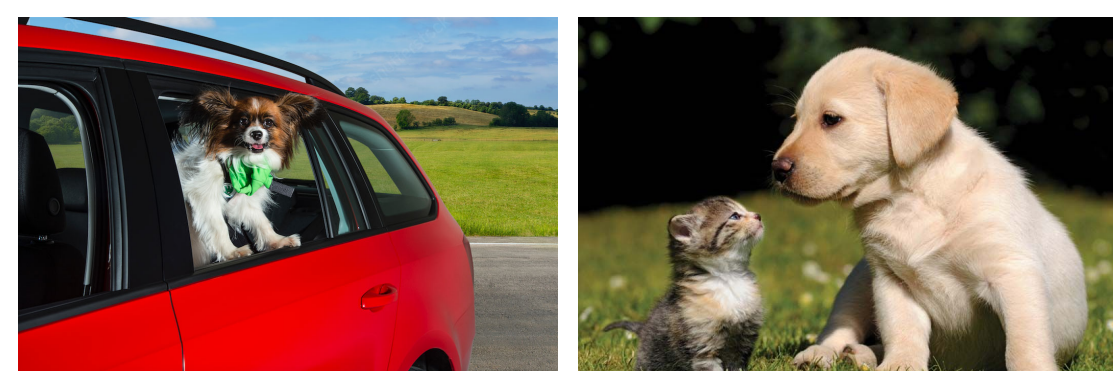
Goal: filter out incorrect answers.

Motivation

Even wrong answers can contain valuable information.

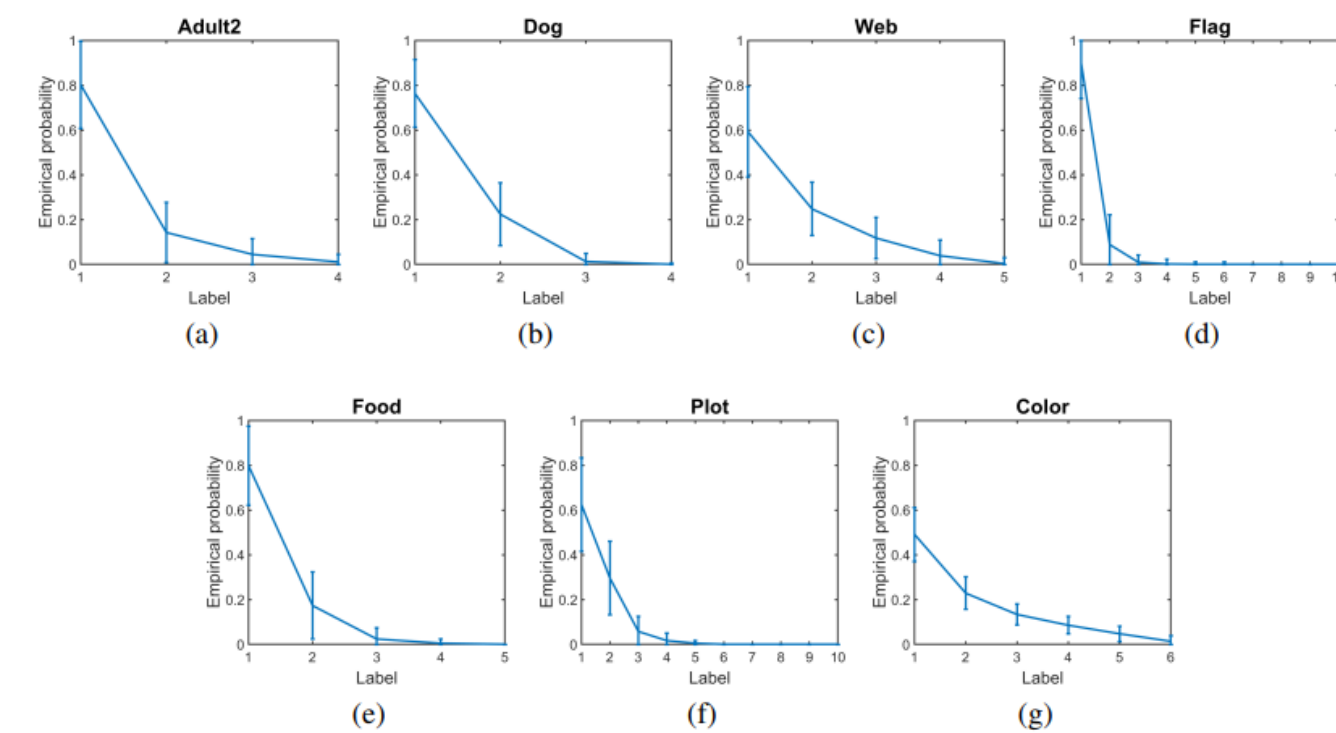
- No ground-truth answer: Preferences, personality
- Multiple correct answers: more than one object in a picture

In CIFAR10H [1] experiments, training with hard labels (only the top-1 answer) achieves a lower accuracy compared to training with top-two labels or full soft labels.



Network	Train accuracy	Test accuracy
VGG-19 (hard)	97.46±0.59%	77.64±1.54%
VGG-19 (top2)	97.00±0.51%	79.20±1.04%
VGG-19 (full)	96.69±0.48%	78.66±0.97%
ResNet18 (hard)	98.47±0.320%	76.49±1.80%
ResNet18 (top2)	98.67±0.491%	80.58±2.36%
ResNet18 (full)	99.19±0.125%	80.93±2.66%

Infer the top two answers and a confusing probability between the two confusing answers!!



The **top two** answers dominate most of the responses in accessible multi-class crowdsourced datasets.

Infer the top two answers and a confusing probability!

Model and Problem setup

- There are n workers and m tasks, i for worker index and j for task index.
- For each task, there exist K possible choices, denoted as $[K]$.
- For each task j , $g_j \in [K]$ denotes the most plausible answer, and $h_j \in [K]$ denotes the second most plausible or confusing answer.
- The data matrix $\mathbf{A} \in \mathbb{R}^{n \times m}$ represents the collected responses.

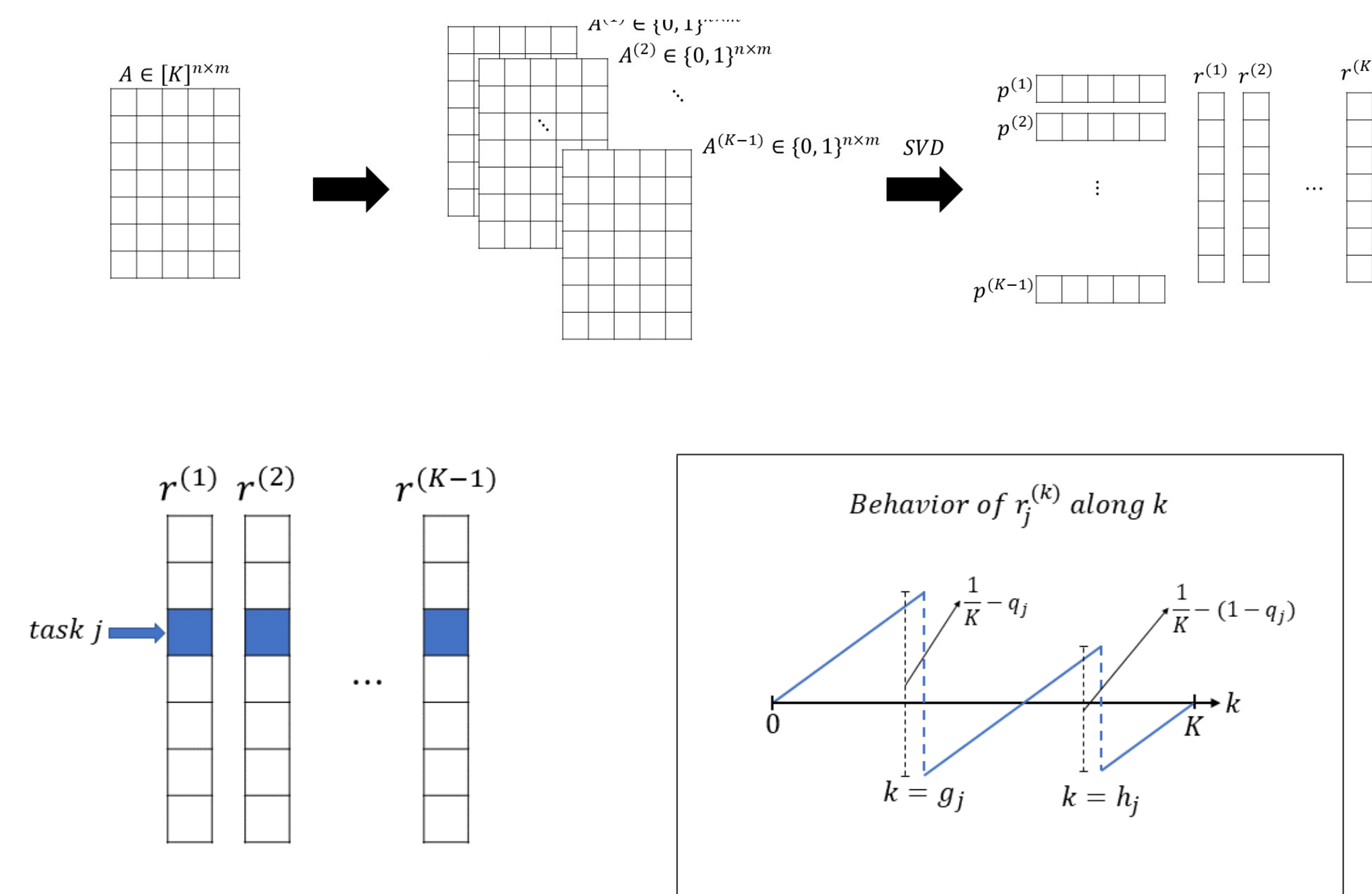
The distribution of A_{ij} is specified by the **worker reliability** $p_i \in [0, 1]$ and **task difficulty** $q_j \in (1/2, 1]$ as follows:

$$A_{ij} = \begin{cases} g_j, & \text{w.p. } s(p_i q_j + \frac{1-p_i}{K}), \\ h_j, & \text{w.p. } s(p_i(1-q_j) + \frac{1-p_i}{K}), \\ b \in [K] \setminus \{g_j, h_j\}, & \text{w.p. } s(\frac{1-p_i}{K}), \\ 0, & \text{w.p. } 1-s. \end{cases}$$

- Each task is assigned to a worker with a probability of s .
- With probability $1 - p_i$, worker i gives a random answer.
- With probability p_i , worker i gives either g_j or h_j .
- q_j is involved in the probability gap between g_j and h_j .

TopTwo1 Algorithm

- Convert a K-ary classification task into K binary classification tasks.
- Decompose each binary matrix $\mathbf{A}^{(k)}$ into $\mathbf{p}^{(k)}$ and $\mathbf{r}^{(k)}$ using SVD.
- For each task j , estimate $\hat{g}_j, \hat{h}_j$ by finding the two largest decreases of $\Delta r_j^{(k)}$ along k .



TopTwo2 Algorithm

- Estimate the worker reliability $\mathbf{p}$ by comparing the worker responses with the estimated top two answers obtained from **TopTwo1**.
- Estimate the task difficulty $\mathbf{q}$ using the $\Delta r_j^{(k)}$ value at $k = \hat{g}_j$.
- Use maximum likelihood estimator(MLE) that maximizes

$$\mathbb{P}(\mathbf{A}|\mathbf{p}, \mathbf{q}, \mathbf{g}, \mathbf{h})$$

which can be simplified as

$$(\hat{g}_j^{\text{MLE}}, \hat{h}_j^{\text{MLE}}) = \arg \max_{a, b \in [K]^2, a \neq b} \sum_{i=1}^n \log \left(\frac{K \hat{p}_i \hat{q}_j}{1 - \hat{p}_i} + 1 \right) \mathbb{I}(A_{ij} = a) + \log \left(\frac{K \hat{p}_i (1 - \hat{q}_j)}{1 - \hat{p}_i} + 1 \right) \mathbb{I}(A_{ij} = b)$$

Performance analysis

Theorem 1. **TopTwo1** guarantees the recovery of (g_j, h_j) with probability at least $1 - \epsilon$ for any tasks with sampling probability $s = \Omega(\frac{\log(K/\epsilon)}{\|\mathbf{p}\|_2^2})$.

$$\mathbb{P}((\hat{g}_j, \hat{h}_j) = (g_j, h_j)) \geq 1 - \epsilon, \quad \text{for all } \forall j \in [m].$$

Also, when $s = o(\frac{\log(K/\epsilon)}{\|\mathbf{p}\|_2^2})$, no algorithm can make the error less than ϵ .

Theorem 2. **TopTwo2** guarantees the recovery of $\{(g_j, h_j)\}_{j=1}^m$ with probability at least $1 - \epsilon$ with sampling probability $s = \Omega(\frac{\log(m/\epsilon)}{nD})$.

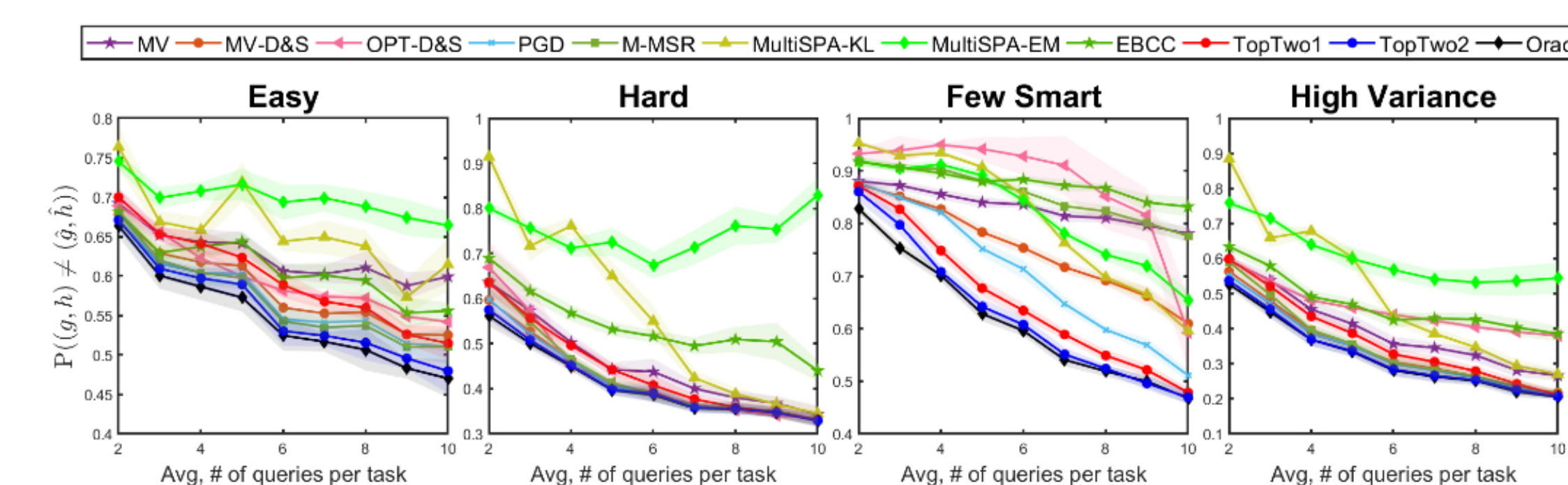
$$\mathbb{P}((\hat{g}_j^{\text{MLE}}, \hat{h}_j^{\text{MLE}}) = (g_j, h_j), \forall j \in [m]) \geq 1 - \epsilon.$$

Also, when $s \leq \frac{1}{nD}$, no algorithm can make the error less than ϵ .

Synthetic experiments

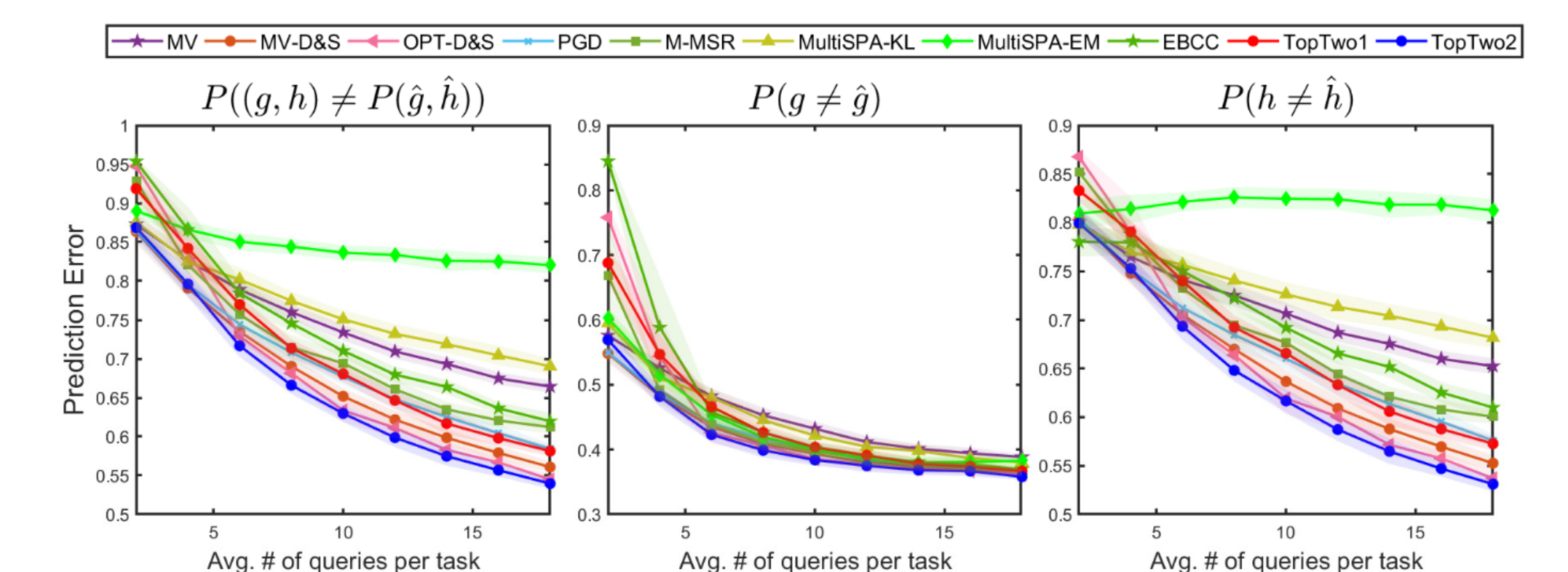
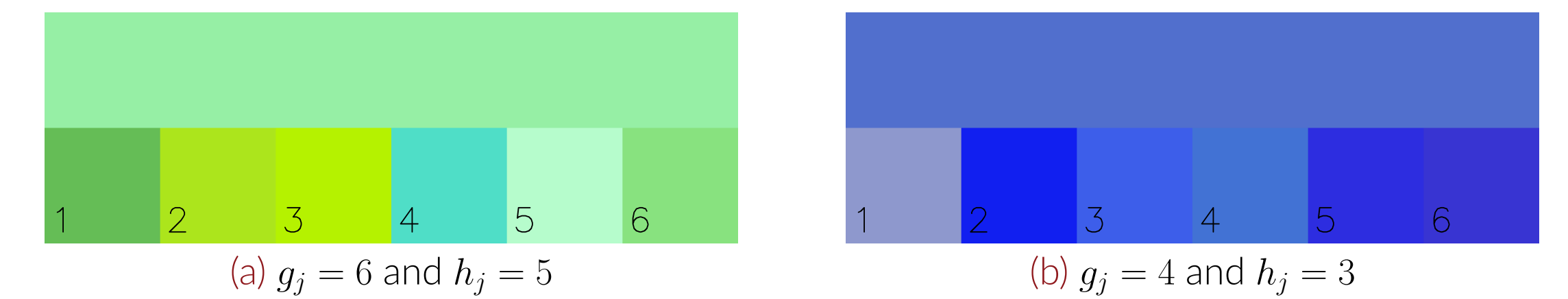
We compare the empirical performance of TopTwo1 and TopTwo2 with other baselines in various scenarios.

	Easy	Hard	Few-smart	High-variance
Worker	$p_i \in [0, 1]$	$p_i \in [0, 1]$	90% $p_i \in [0, 0.1]$ 10% $p_i \in [0.9, 1]$	$p_i \in [0, 1]$
Task	$q_j \in [0.9, 1]$	$q_j \in [0.5, 0.6]$	$q_j \in [0.5, 1]$	50% $q_j \in [0.5, 0.6]$ 50% $q_j \in [0.9, 1.0]$

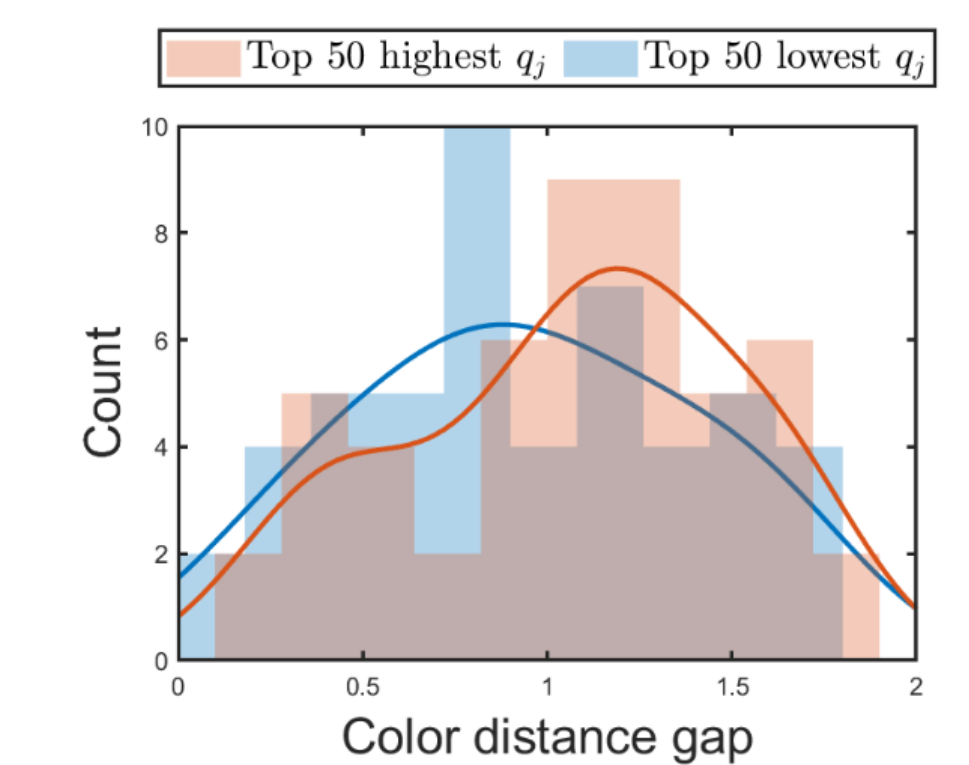


Real Experiments

For the real experiments, we designed a **color** comparison task where we asked the crowd to choose a color from six given choices, that looks the most similar to a reference color of each task.



TopTwo2 achieve the best performance in detecting (g_j, h_j) in all ranges.



The difficult group (blue, having lowest q_j) have a smaller distance gap than those of the easy task group(red, having highest q_j).

Our algorithm can identify difficult tasks in real datasets.

Conclusion

- We propose a top-two model for multi-choice crowdsourcing, and two algorithms for inferring the top-two answers.
- We present the efficiency of our algorithm through theoretical analysis and experiments on both synthetic and real datasets.

References

- [1] Joshua C. Peterson, Ruairidh M. Battleday, Thomas L. Griffiths, and Olga Russakovsky. Human uncertainty makes classification more robust, 2019.